

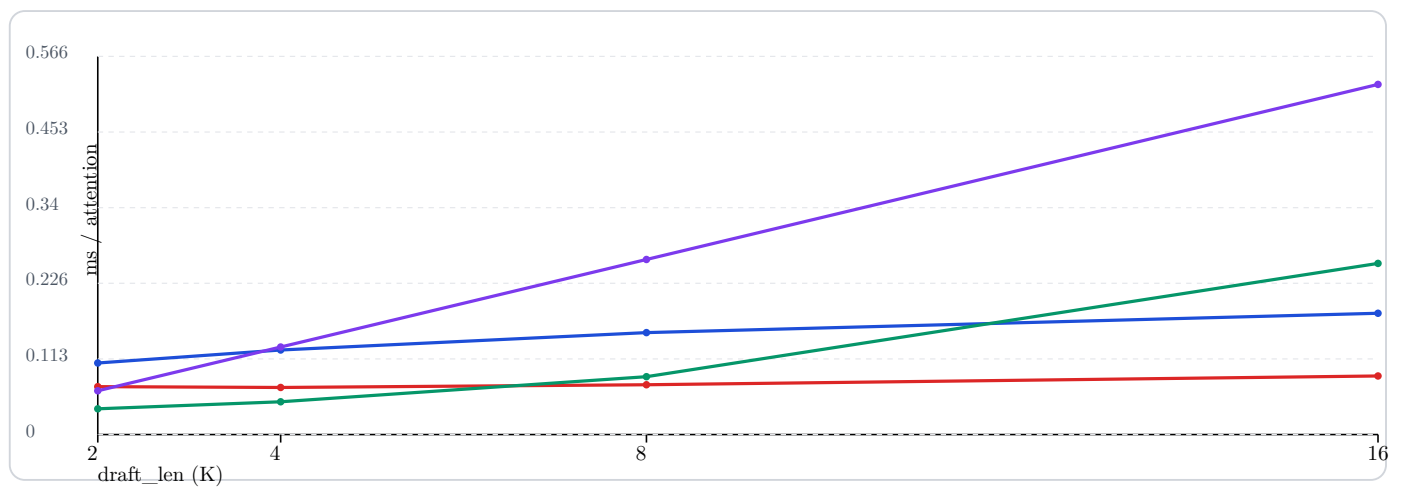
A40 Decode Attention Speed (Final)

Runs: A40 spot, bf16, implementation=cudnn, prefix_len=1024, iterations=100, repeats=5, batch_size=1.

Pallas series is intentionally omitted from these plots to keep the non-pallas curves readable on a single scale. AR baseline is measured at batch=1, query_len=1 and scaled by K for each draft length point.

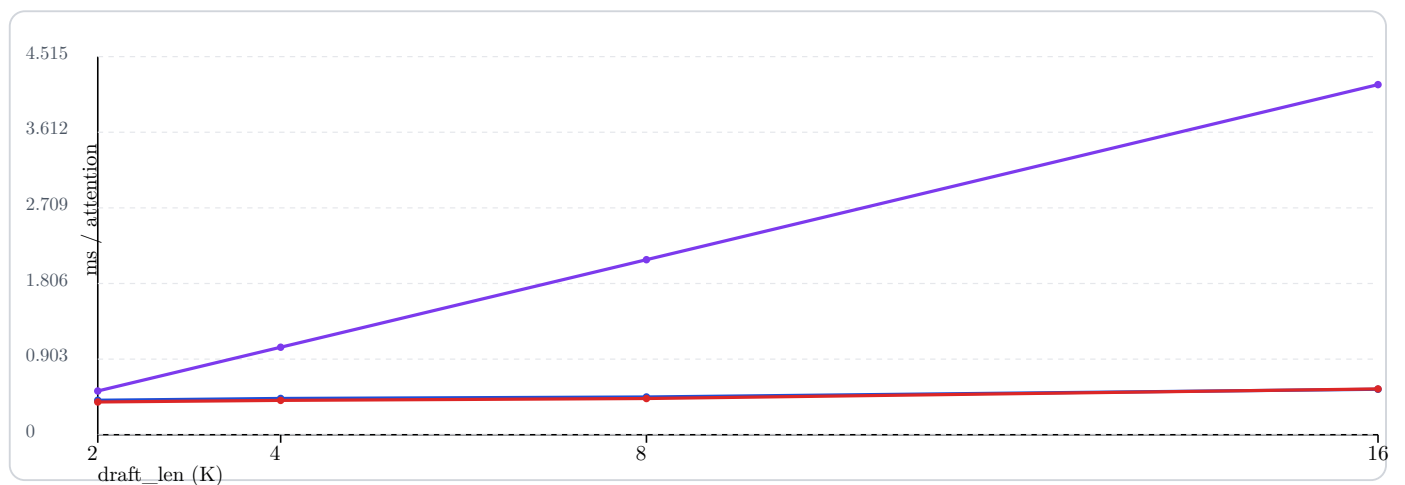
1B-style MHA (32/32, head_dim 64) - Kernel terms

- kernel structured
- kernel dense
- kernel dense+jax-zero
- AR baseline (K x len1)



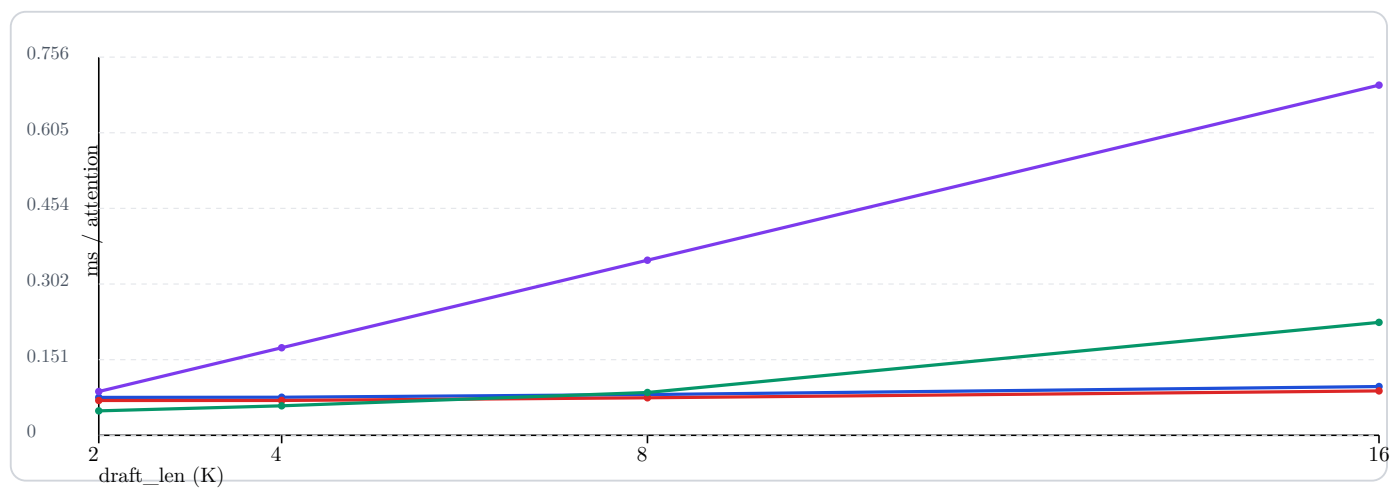
1B-style MHA (32/32, head_dim 64) - Native decode

- native structured
- native dense
- AR baseline (K x len1)



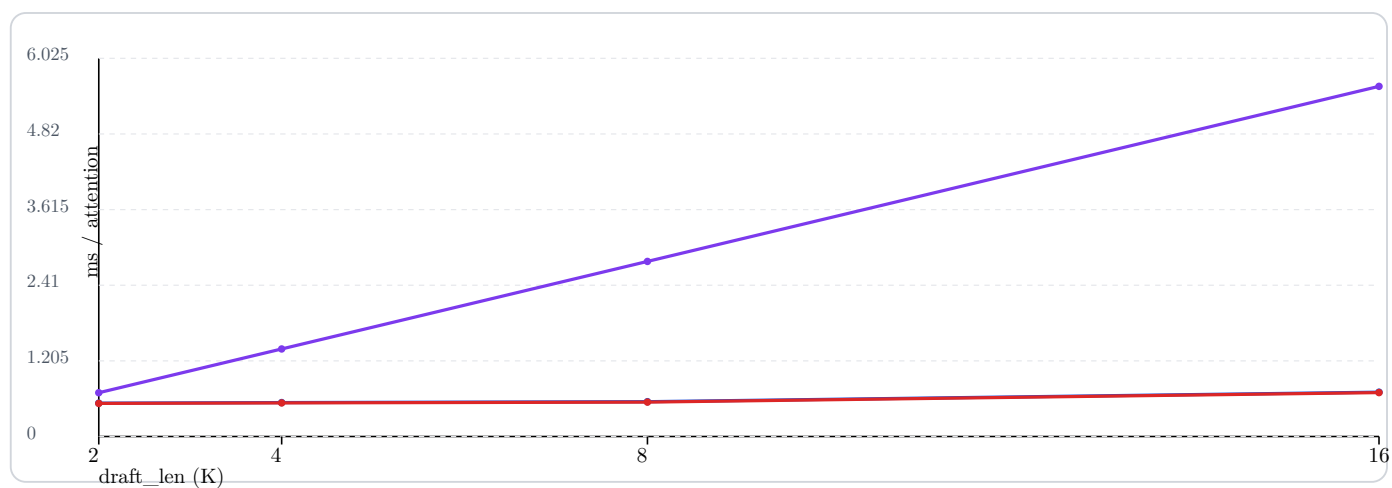
3B-style GQA (24/8, head_dim 128) - Kernel terms

- kernel structured
- kernel dense
- kernel dense+jax-zero
- AR baseline (K x len1)



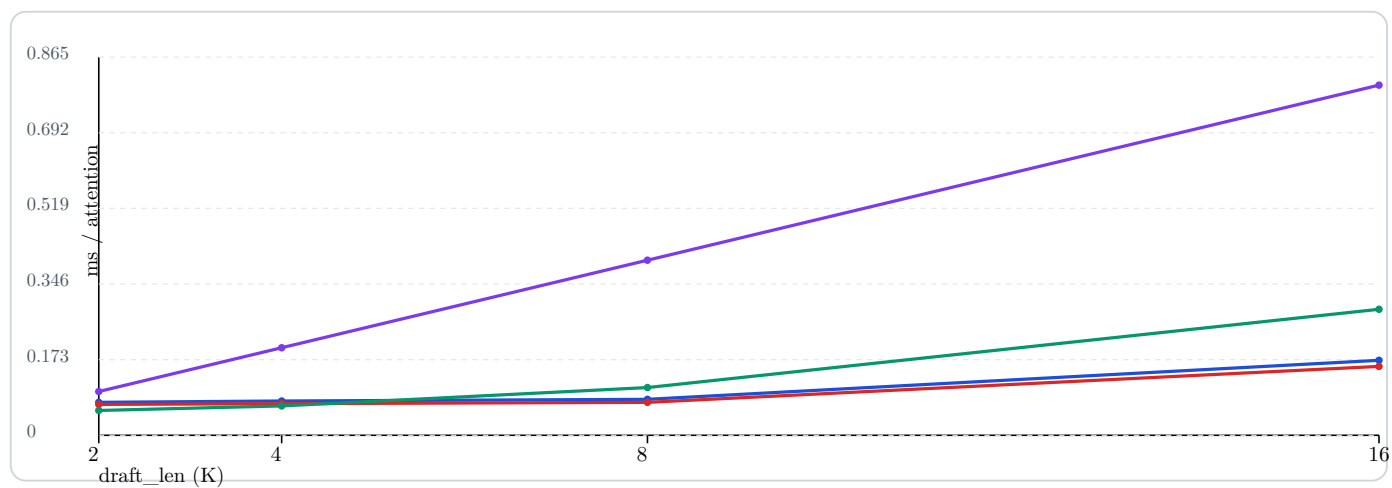
3B-style GQA (24/8, head_dim 128) - Native decode

- native structured
- native dense
- AR baseline (K x len1)



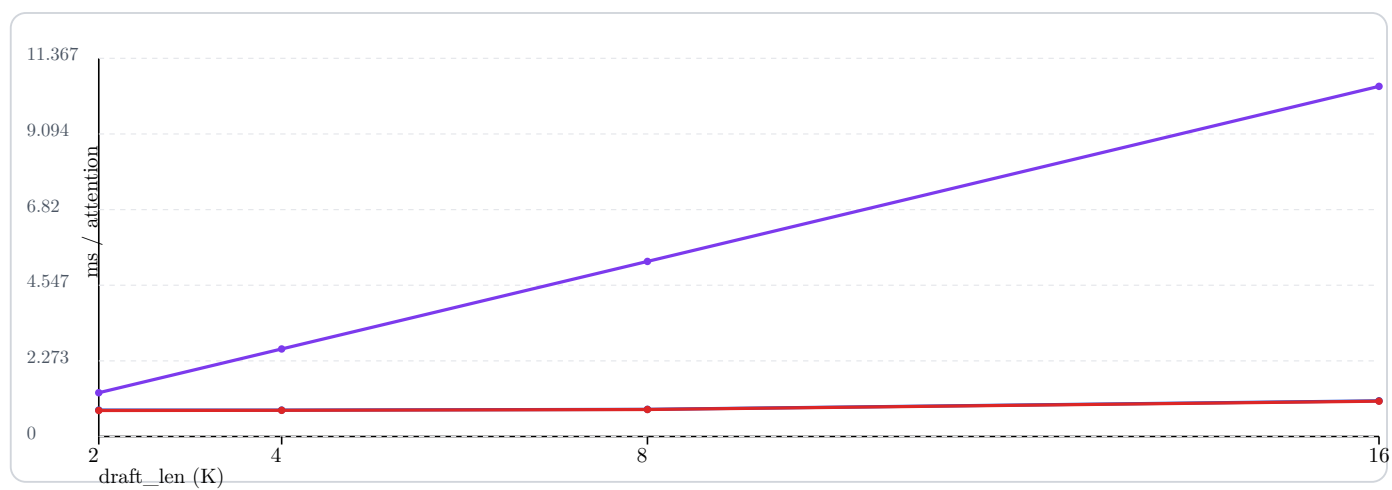
7B-style GQA (32/8, head_dim 128) - Kernel terms

- kernel structured
- kernel dense
- kernel dense+jax-zero
- AR baseline (K x len1)



7B-style GQA (32/8, head_dim 128) - Native decode

- native structured
- native dense
- AR baseline (K x len1)

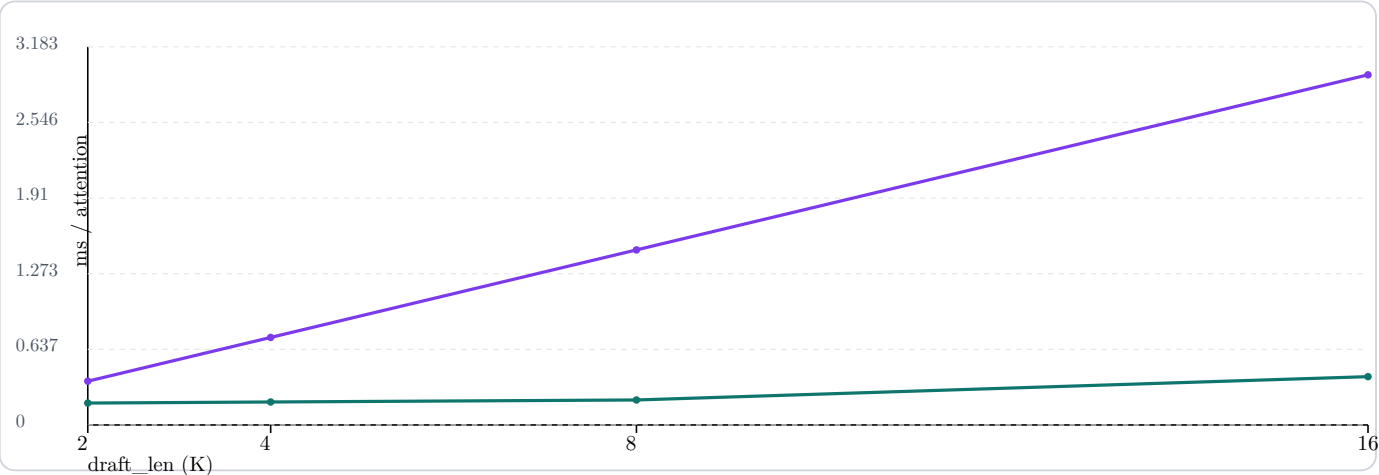


A40 MLP Throughput — AR vs TiDAR Tokens

This section compares MLP-only cost using SwiGLU projections. AR baseline is measured on `len=1` and scaled by `K`, while TiDAR runs the full $L = K + K^2$ token block in one pass.

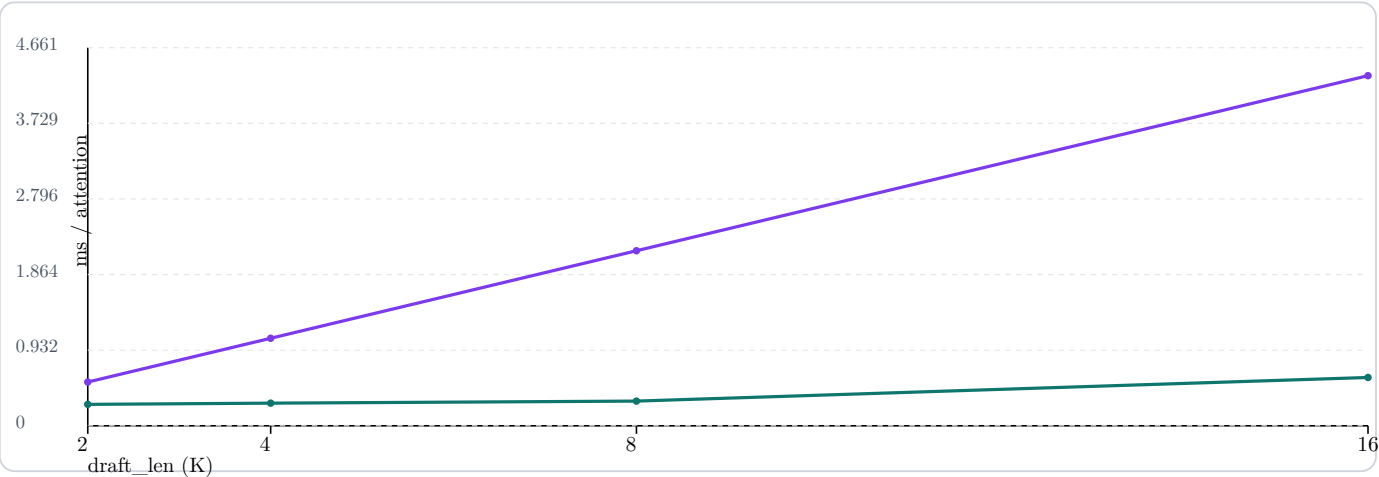
1B-style MHA (`d_model=2048`, `d_ff=8192`) - MLP only

- TiDAR MLP at $L=K+K^2$
- AR MLP baseline ($K \times len1$)



3B-style GQA (`d_model=3072`, `d_ff=8192`) - MLP only

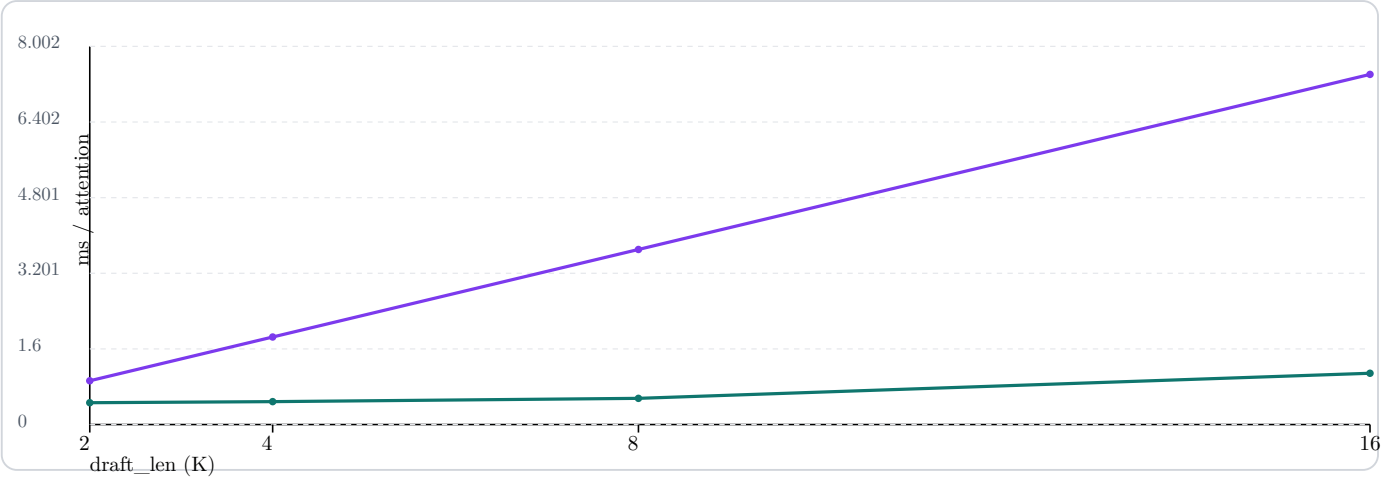
- TiDAR MLP at $L=K+K^2$
- AR MLP baseline ($K \times len1$)



7B-style GQA (d_model=4096, d_ff=11008) - MLP only

TiDAR MLP at $L=K+K^2$

AR MLP baseline ($K \times \text{len1}$)



A40 Greedy Decode: Staged Softmax vs Greedy No-Softmax

This section separates two different measurements to avoid mixing signals:

- End-to-end full decode pass (includes full model forward cost).
- Isolated sampling microbenchmark (focuses on candidate sampling/rejection/commit path only).

End-to-end full decode (A40, draft_len=8, prefill_tokens=1024, single pass)

These gains are very small and near noise floor for one-pass runs:

- 500m: 62.0088 -> 62.0902 tokens/s (+0.131%)
- 1b: 210.7911 -> 210.8543 tokens/s (+0.030%)
- 3b: 63.2153 -> 63.2564 tokens/s (+0.065%)

Mean speedup across these three runs: +0.075%.

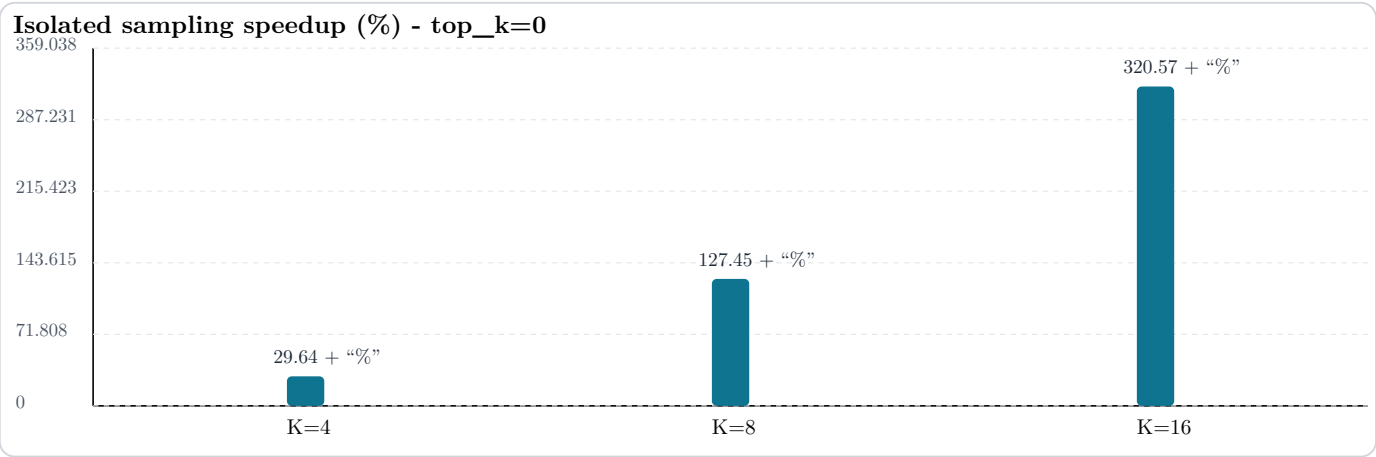
Isolated sampling microbenchmark (not full forward pass)

Benchmark definition:

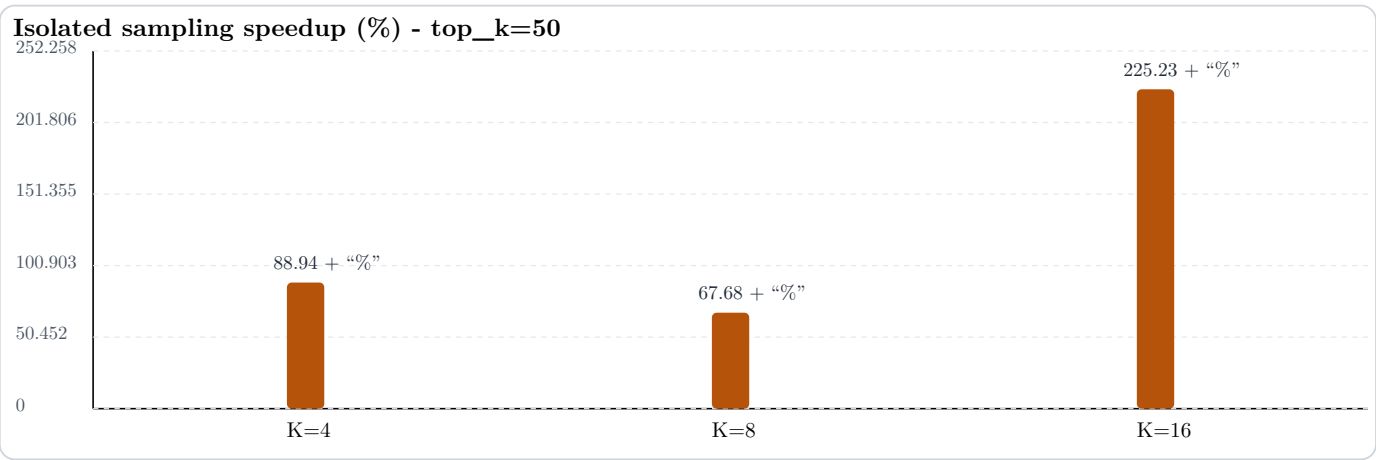
- Old path: sample K (verify) + sample K^2 (all predraft rows) + determine row + commit.
- New path: sample K (verify) + determine row + sample selected K row only + commit.

The following deltas are from this isolated sampling path and are not end-to-end decode speedups.

■ new path speedup over old path



■ new path speedup over old path



A40 Isolated Sampling: AR (B=1) vs TiDAR by K

This page isolates only the sampling/rejection path (no transformer forward) and compares:

- AR baseline sampled at `batch=1`, `len=1`, then scaled by K ($K \times \text{len1}$) for parity with TiDAR committed-token span.
- TiDAR staged path (sample verify, reject/select, sample only selected predraft row).
- Single fixed setup for this sweep: `prefix_len=1024`, vocab 49152, A40.
- Variants: greedy (`temperature=0`, `top_k=0`) and non-greedy (`temperature=1`, `top_k=50`).

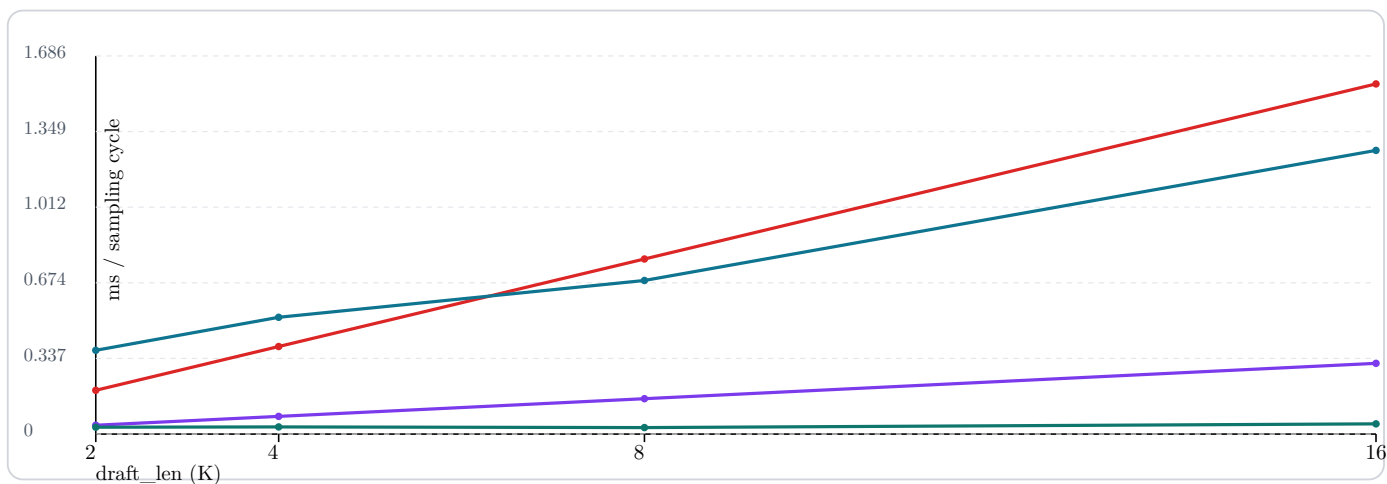
500m sampling-only: AR scaled vs TiDAR staged

■ AR scaled ($K \times \text{len1}$), greedy

■ TiDAR staged sampling, greedy

■ AR scaled ($K \times \text{len1}$), non-greedy `top_k=50`

■ TiDAR staged sampling, non-greedy `top_k=50`



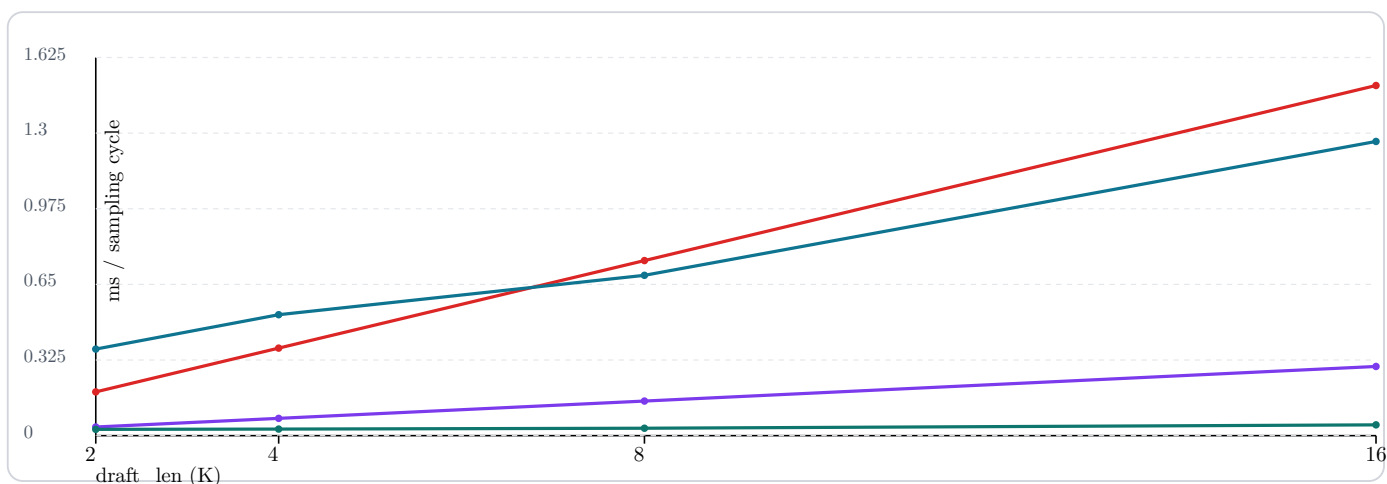
1b sampling-only: AR scaled vs TiDAR staged

■ AR scaled ($K \times \text{len1}$), greedy

■ TiDAR staged sampling, greedy

■ AR scaled ($K \times \text{len1}$), non-greedy `top_k=50`

■ TiDAR staged sampling, non-greedy `top_k=50`



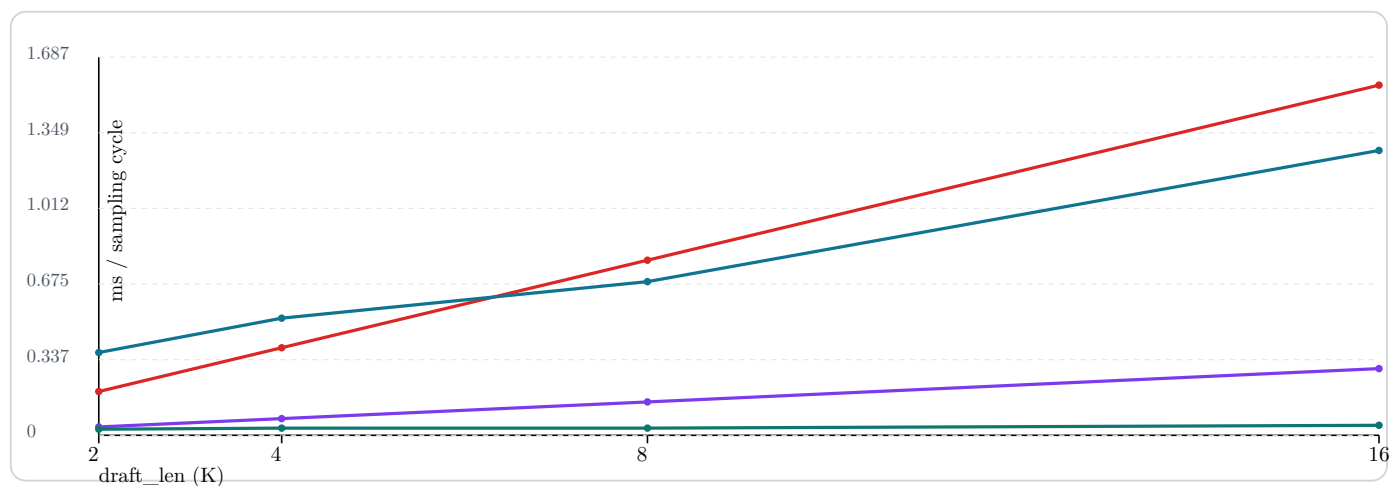
3b sampling-only: AR scaled vs TiDAR staged

AR scaled ($K \times \text{len1}$), greedy

TiDAR staged sampling, greedy

AR scaled ($K \times \text{len1}$), non-greedy top_k=50

TiDAR staged sampling, non-greedy top_k=50



Greedy curves show strong TiDAR sampling advantage as K grows. In non-greedy top_k=50, TiDAR crosses over from slower at low K (2,4) to faster at higher K (8,16).